

Periodic Table	2
Matter	2
Atoms, Molecules, Isotopes, and Ions	3
Bonds	4
Chemical Reactions	4
Naming Molecular Compounds	5
Naming Ionic Compounds	5
Polyatomic Ions	6
Ion Symbols	6
Naming Acids	6

Periodic Table

Periodic Law	elements' properties are functions of their atomic numbers
Periods/Series	rows in the periodic table
Groups	columns in the periodic table
Alkali Metals	all elements in group 1, excluding hydrogen valance shell has 1 electron
Alkaline Earth Metals	all elements in group 2 these elements form compounds consisting of 2 atoms and 1 hydrogen valence shell has 2 electrons
Main-Group Elements/ Representative Elements	elements in groups 1, 2, 13-18
Transition Metas	elements in groups 3-12
Pnictogens	elements in group 15 valence shell has 5 electrons
Chalcogens	elements in group 16 valence shell has 6 electrons
Halogens	elements in group 17 valence shell has 7 electrons
Noble Gasses	elements in group 18 valence shell has 8 electrons
Inner Transition Metals	elements in the bottom of period 3, in the last two groups its usually listed as a separate table
Lanthanides	elements in period 1 of inner transition metals
Actinides	elements in period 2 of the inner transition metals
Metal	shiny, malleable, and are good heat & electricity conductors
Metalloid	possess some metal properties and are moderate heat & electricity conductors
Nonmetal	dull and are poor heat & electricity conductors

Matter

Matter	a pure substance, if all particles are identical a mixture if there are more than two types of particles present
Particle	a physical unit of matter
Element	a pure substance made of one atom

Atoms, Molecules, Isotopes, and Ions

Atom	a singular, electrically-neutral unit of an element
Atomic	referring to the atom as a whole
Atomic Number	an element's number on the periodic table that corresponds to the number of protons, neutrons, and electrons in an atom
Mass Number	sum of protons and neutrons
Subatomic	referring to particles within the atom
Proton	positively-charged subatomic particle
Neutron	neutrally-charged subatomic particle quantity does not change
Electron	negatively-charged subatomic particle
Electron Shell	layer of electrons revolving around the nucleus all shells hold eight electrons total, except the first, which holds two
Valence Electrons	the outermost electron shell
Nucleus	the atom's core, contains protons and neutrons
Orbials	the area surrounding the nucleus, each containing 2 electrons
Isotope	elements that have lost or gained neutrons, $X - \#$ they have the same atomic numbers but different mass numbers
Ion	elements that have lost or gained electrons, making them electrically charged
Cation	positively-charged from losing electrons
Anion	negatively-charged from gaining electrons
Electron Transfer	atoms becoming ions through electron exchange

Bonds

Compound	pure substances made of two or more atoms
Chemical Bond	connection between two or more atoms
Electronegativity	the atom's ability to attract electrons from other atoms the Pauling Scale is the easiest way to determine this
Molecule	two or more atoms
Acids	compounds containing hydrogen
Ionic Bonds	multiple ions transferring electrons occur between at least one nonmetal and optionally a metalloid electronegativity difference does exceed 1.7
Covalent Bonds	multiple atoms sharing at least one electron pair can occur between metals electronegativity difference does not exceed 1.7
Polar Covalent Bonds	atoms have a electronegativity difference $> .4$ negative charge is not evenly distributed, causing a dipole moment creates slightly-positive (δ^+) and negative (δ^-) ends
Dipole Moment	one end of a polar covalent bond is positive, and the other is negative
Non Polar Covalent Bonds	atoms have the electronegativity difference $< .4$ electrons are shared evenly
van der Waals Forces	intermolecular forces that do not involve covalent bonds or ions
Hydrogen Bonds	hydrogen will be temporarily attracted to any neighboring negative charge individually, very weak; collectively, very strong
London Dispersion Forces	weak temporary attractions between atoms or molecules free-floating atoms or molecules in a cluster will be attracted to each other if they have partial charge
Solution	
Solvent	

Chemical Reactions

exothermic	releases energy	
endothermic	require energy	
Synthesis	an exothermic reaction occurring when two reactants form one product	$A + B \rightarrow AB$

Decomposition	an endothermic reaction occurring when reactant breaks down into smaller products	$AB \rightarrow A + B$
Single Replacement	one reactant replaces part of the other reactant compound	$AB + C \rightarrow AC + B$
Double Replacement		$AB + CD \rightarrow AC + BD$
Precipitation		
Acid Base		
Combustion		

Naming Molecular Compounds

mono-		1
di-		2
tri-		3
tetra-		4
penta-		5
hexa-		6
hepta-		7
octo-		8
nona-		9
deca-		10
hydro-	hydro acid	
-ide	last element in the compound	

Naming Ionic Compounds

-ic	positive ions/hydro acid/oxo acid more oxygen atoms
-ide	monatomic ion

-ous	oxo acid
-[h]ol	alcohol
(I)	metal with a +1 charge
(II)	metal with a +2 charge
(III)	metal with a +3 charge
(IV)	metal with a +4 charge

Polyatomic Ions

1+	-ium (metal)
1-	-ide (nonmetal)
2-	-ite (nonmetal) -ate (nonmetal with largest number of oxygen atoms)

- cations and anions can form ternary ionic compounds
- the name uses both, with the cation being named first

Ion Symbols

Superscript	number of valence electrons and charge	$X^{\pm\#}$
Subscript	number of ions	$X_{\#}$

Naming Acids